

ADRC hands-on workshop for single-nuclei RNA-seq (Week 2)

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August 3, 2026

Resources for this workshop

- GitHub: https://github.com/linnykos/ADRC_workshop_2026
 - This is where the code we showcase will be deposited
 - Has the Google Slide link to these slides
- BIOST 545 lecture notes: <https://linnykos.github.io/scOmicNotes/>
 - Lecture notes written by Kevin that go more into the conceptual understanding of single-cell methods and its usage in different omics
- Primer overview of sc/snRNAseq analysis and decisions:
 - <https://onlinelibrary.wiley.com/doi/10.1002/glia.24633>

Announcements

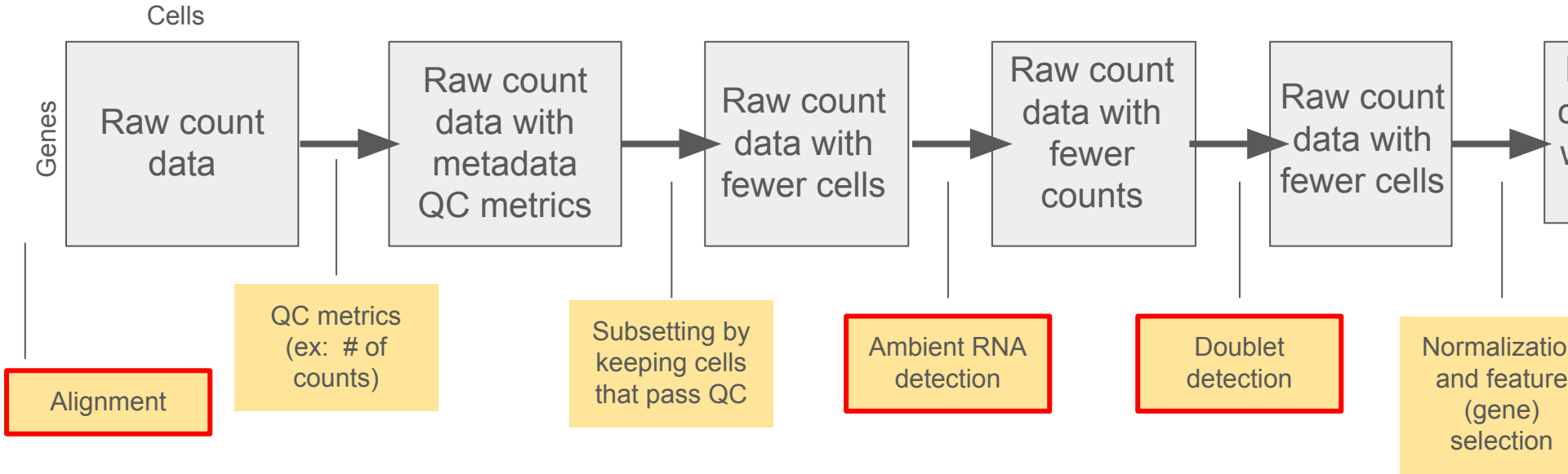
- Next week, Katie (Prater) won't be here, so it'll just be Kevin :)
- We're slightly changing the order of content
 - ~~Week 1: The high-level workflow~~
 - Week 2: Sequence alignment, ambient RNA removal, doublet detection (Today)
 - Week 3: Normalizing gene expression counts, selecting genes to analyze, starting batch correction
 - Week 4: Batch correction, clustering, differential expression,
 - Week 5: Cell-type labeling, bulk deconvolution

Questions from week 1?

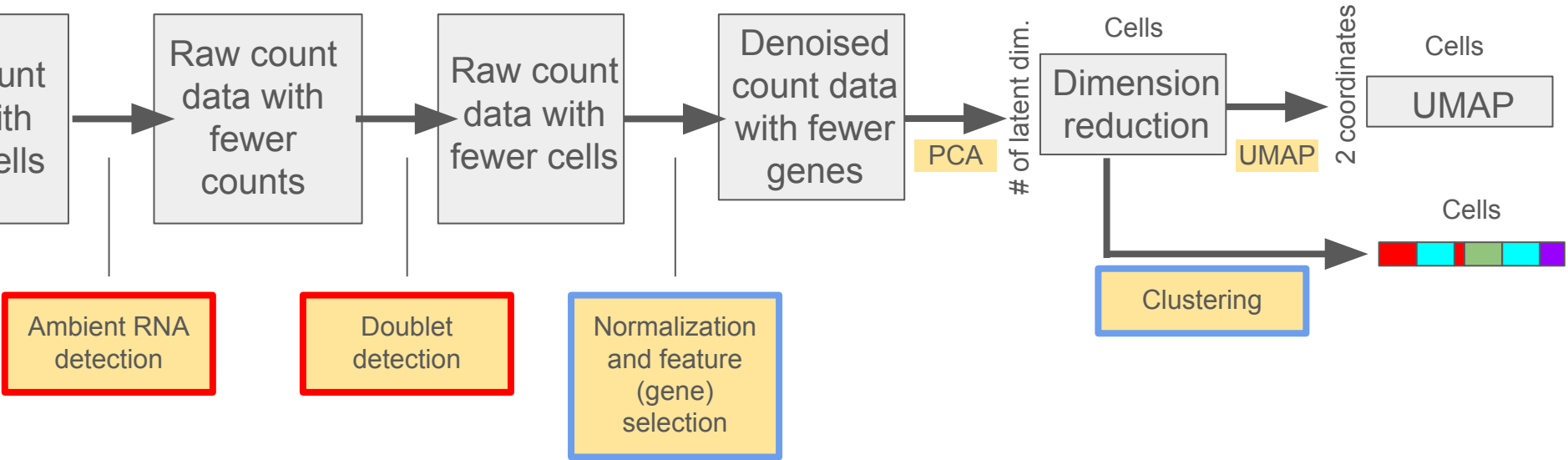
Any questions for the group to discuss (not specific to your project)?

The recorded content for Week 2

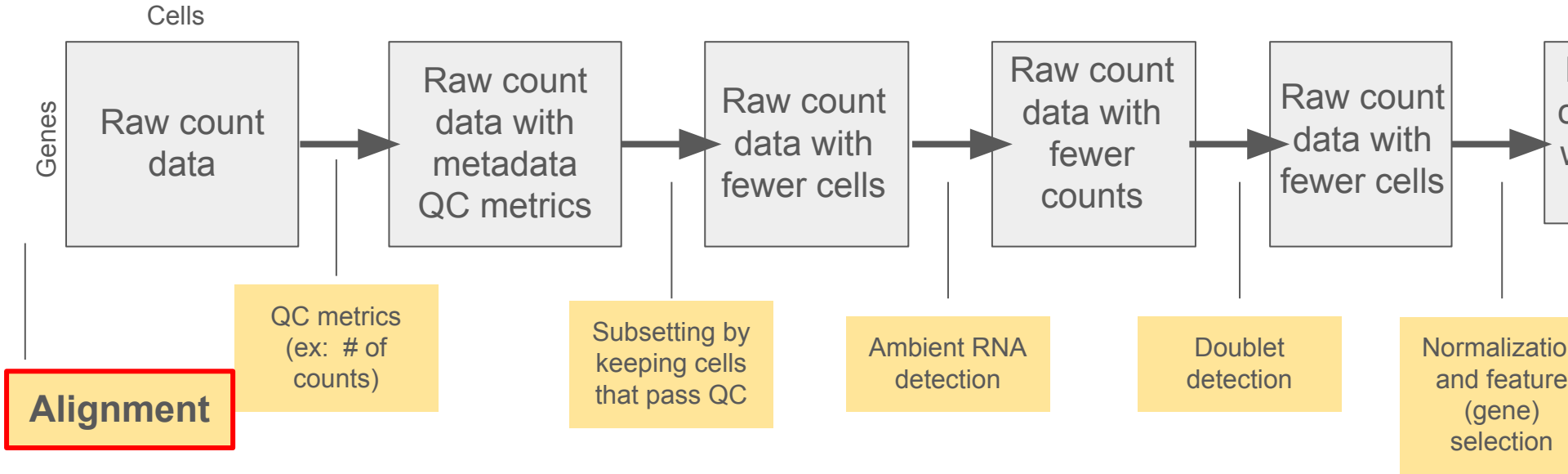
How we're going to be adjusting the workflow today:



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Alignment: The step in your workflow you potentially never think about!

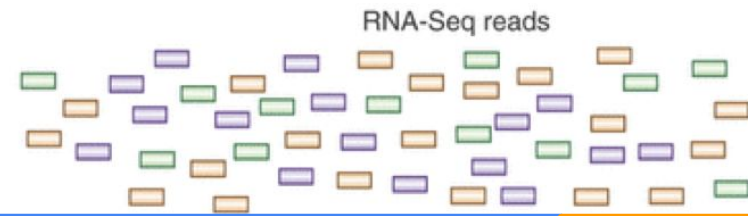


What is alignment?

- Taking the sequence reads from your dataset and “aligning” them to the genome of choice
- This assigns each read to a gene with some probability of accuracy
- The assigned counts are what then make up your “counts matrix”

For a short-ish tutorial on STAR and alignment you can read here:

https://hbctraining.github.io/Intro-to-rnaseq-hpc-orchestra/lessons/07_rnaseq_workflow.html

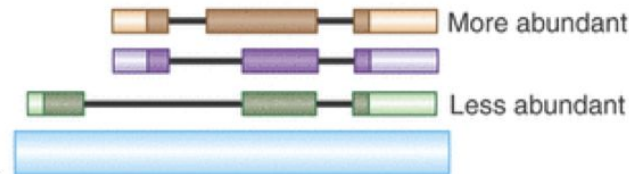


**When you have
a genome**

Align reads to
genome



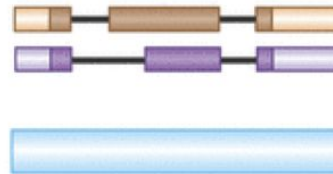
Assemble transcripts
from spliced alignments



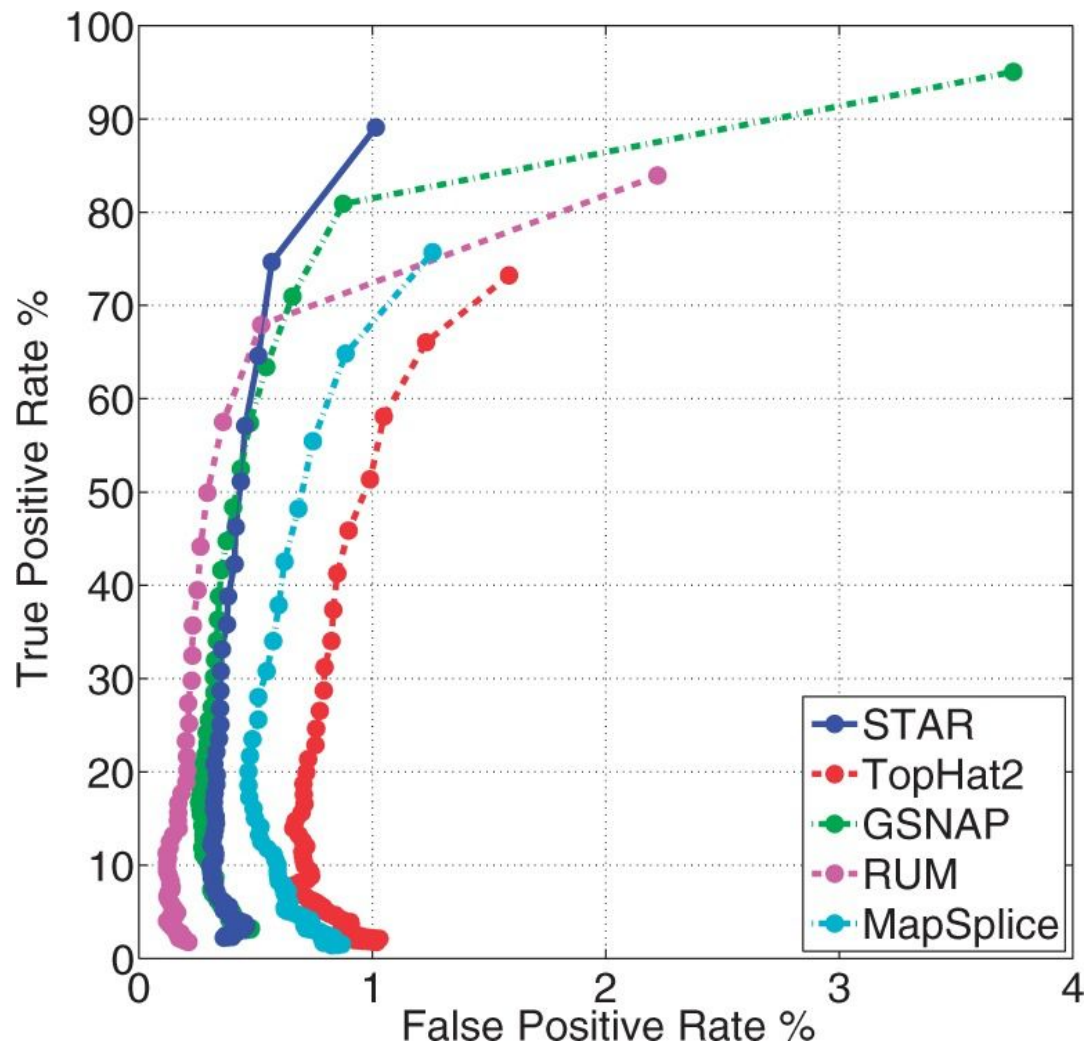
Assemble transcripts
de novo



Align transcripts to
genome



**When you
don't
have a
genome**

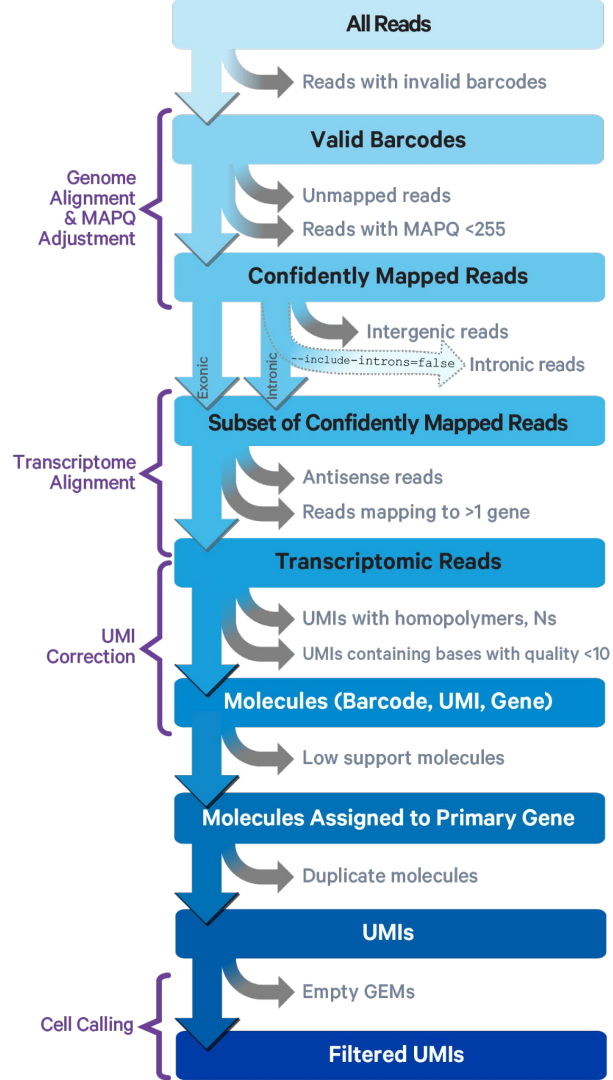


Dobin A, Davis CA, Schlesinger F, Drenkow J, Zaleski C, Jha S, Batut P, Chaisson M, Gingeras TR. STAR: ultrafast universal RNA-seq aligner. *Bioinformatics*. 2013 Jan 1;29(1):15-21. doi: 10.1093/bioinformatics/bts635. Epub 2012 Oct 25. PMID: 23104886; PMCID: PMC3530905.



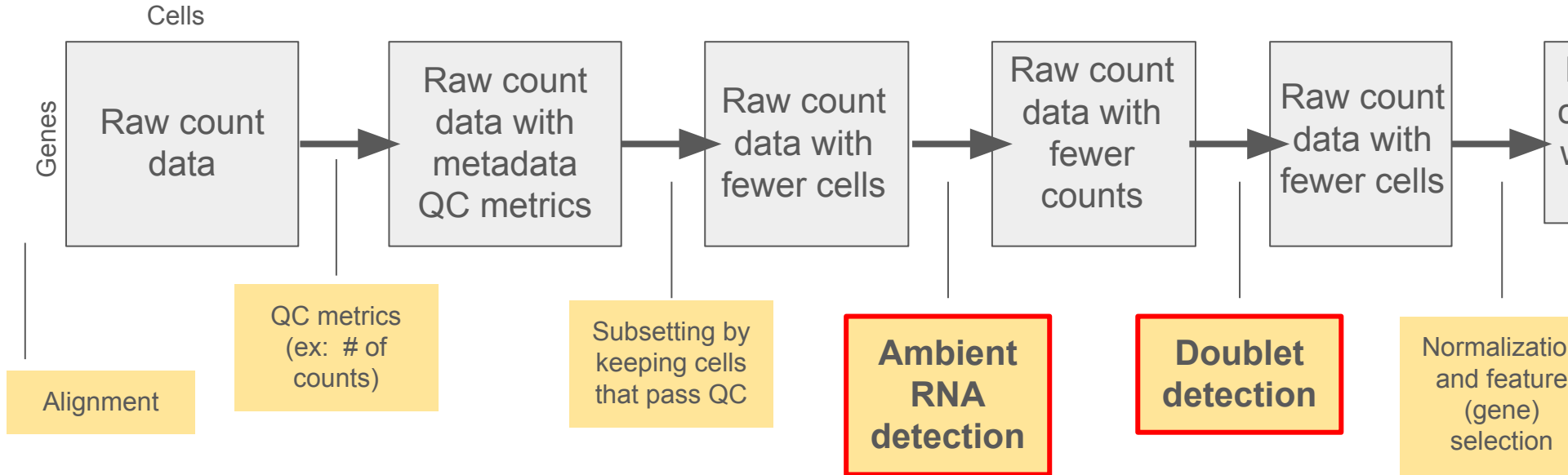
Mapq (mapping quality)
Score

10X CellRanger

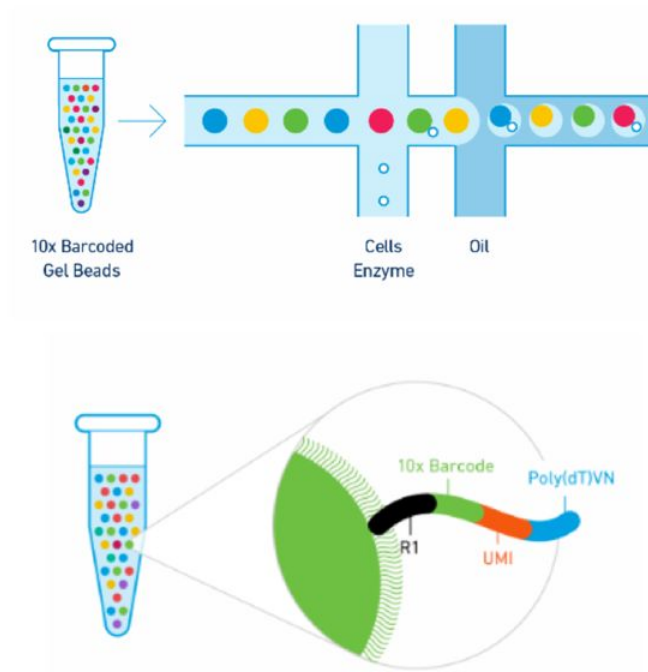


Start from the non-filtered reads to do ambient RNA adjustment

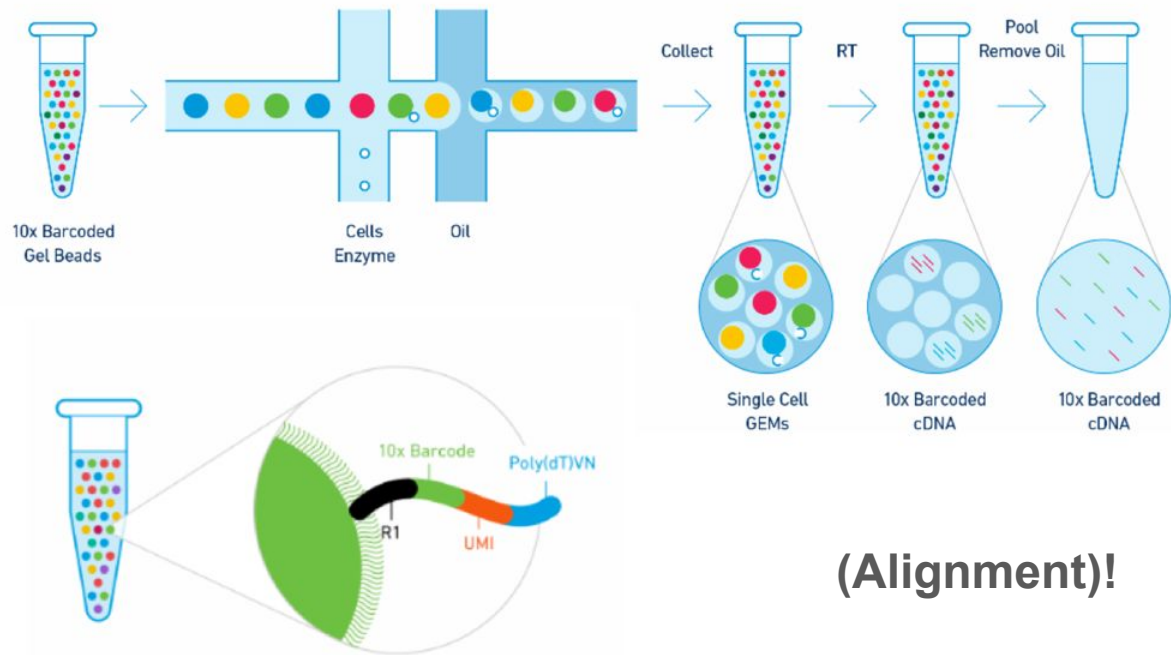
Correcting your data for technical artifacts: Ambient RNA and Doublets



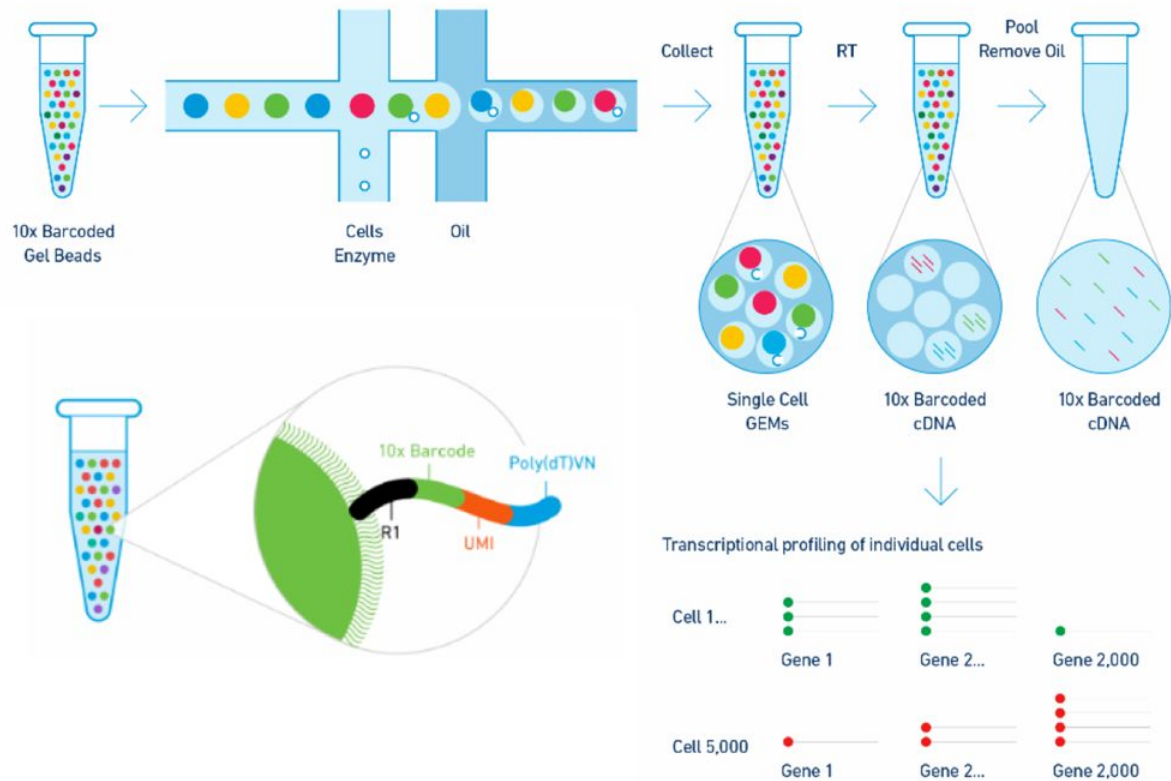
10X Genomics for single nuclei isolation



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This process generates two forms of technical artifacts:

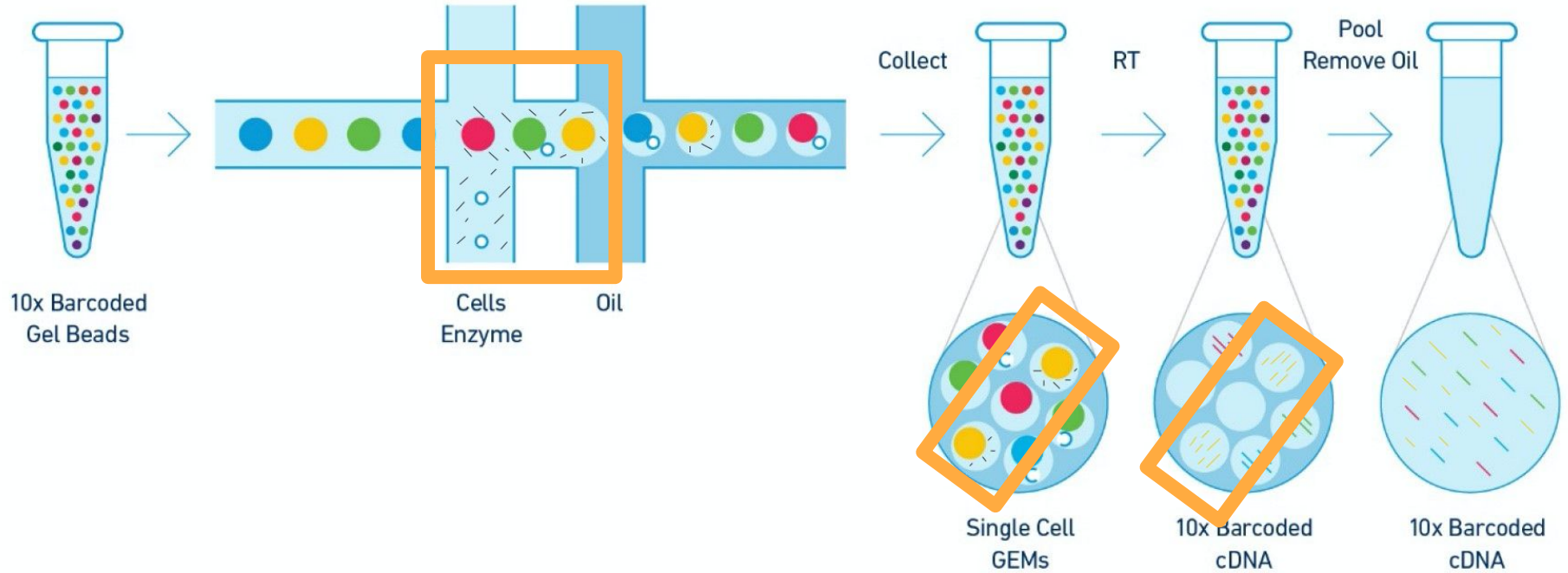
Ambient RNA

&

Doublets

(Note that if you use a different form of single-cell parcellation i.e. patch-seq or multi-well indexing, one or both of these may be less applicable.)

Ambient RNA



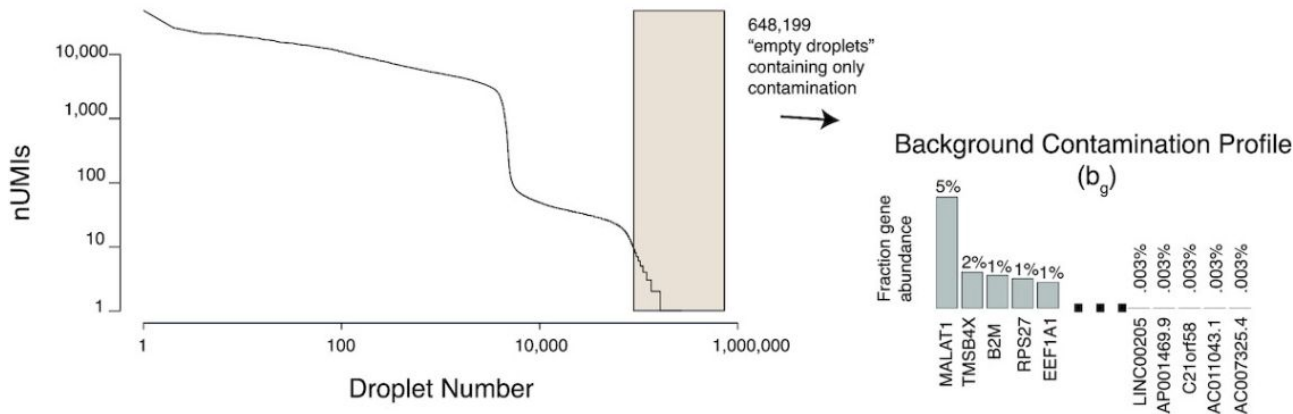
Ambient RNA detection

- In R: `install.packages('SoupX')`
 - See: <https://github.com/constantAmateur/SoupX>
 - See more in <https://rawcdn.githack.com/constantAmateur/SoupX/204b602418df12e9fdb4b68775a8b486c6504fe4/inst/doc/pbmcTutorial.html>
- Warning: Ambient RNA detection works better when you provide it cell clusters. Stay tuned for when we discuss how to cluster cells. If you have a dataset that is unclustered, you can talk to us, or use provided clusters for your working time today.
- Read more: Young and Behjati, Giga Science, 2020
<https://academic.oup.com/gigascience/article/9/12/giaa151/6049831>

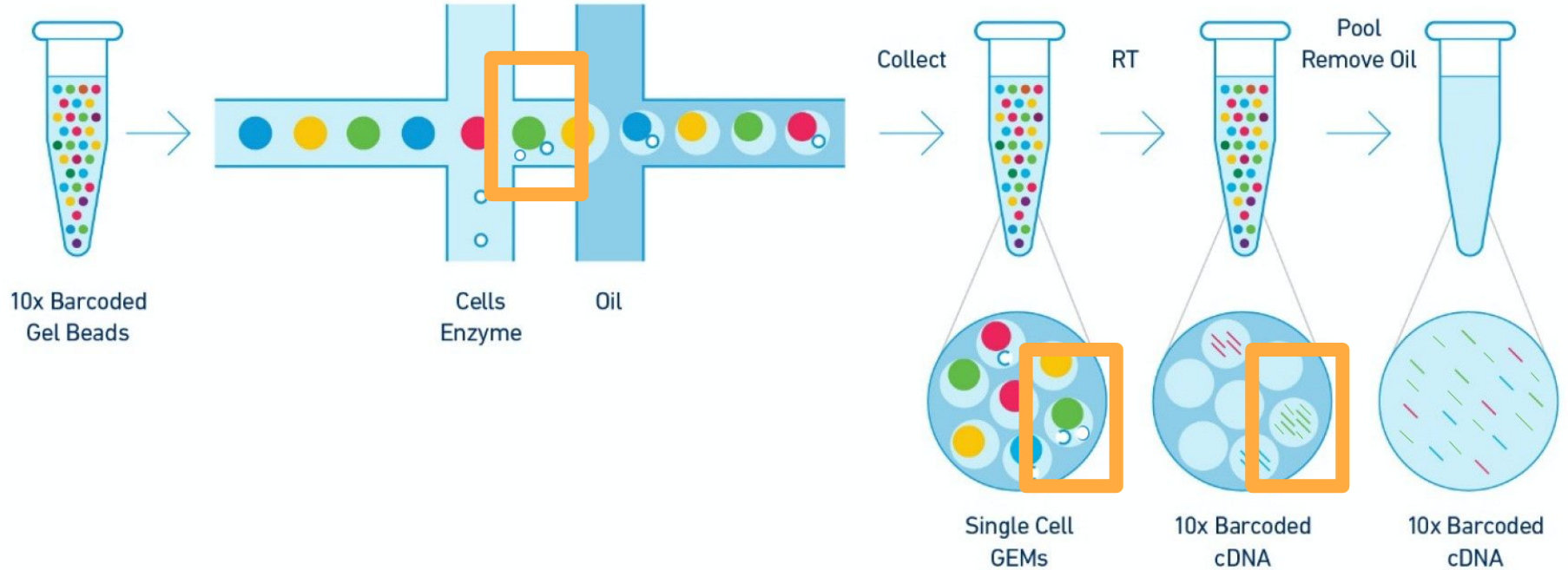
SoupX: 1 sentence summary

- Estimates the ambient (“soup”) RNA profile from empty droplets, infers each cell’s contamination fraction using genes that should be unexpressed in those cells, and subtracts this inferred background to return a decontaminated count matrix for droplet-based scRNA-seq data

1. Determine the expression profile of contamination



Doublets

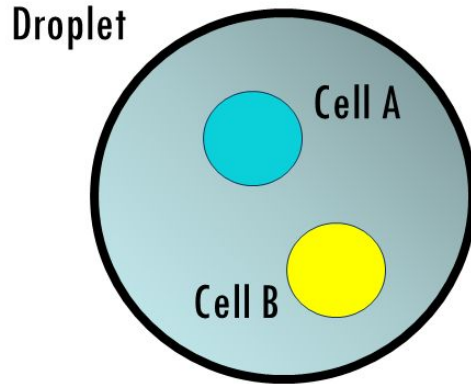


Doublet detection

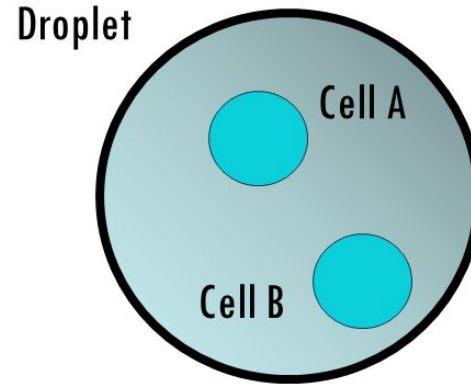
- We'll be using scDblFinder (Germain et al., F1000Res., 2022, <https://pmc.ncbi.nlm.nih.gov/articles/PMC9204188/>), as suggested by Heumos et al. (Nat Rev Genet., 2023).
- This requires us to describe how to convert a Seurat object into a SingleCellExperiment (see more in https://satijalab.org/seurat/archive/v3.1/conversion_vignette)
- In R: `BiocManager::install("scDblFinder")`
 - See: <https://bioconductor.org/packages/release/bioc/html/scDblFinder.html>
 - See more in <https://bioconductor.org/books/3.15/OSCA.advanced/doublet-detection.html>
- The code we'll be using: https://github.com/linnykos/ADRC_workshop_2025/blob/main/Day1/scDblFinder.R

Different types of doublets

Heterotypic doublet

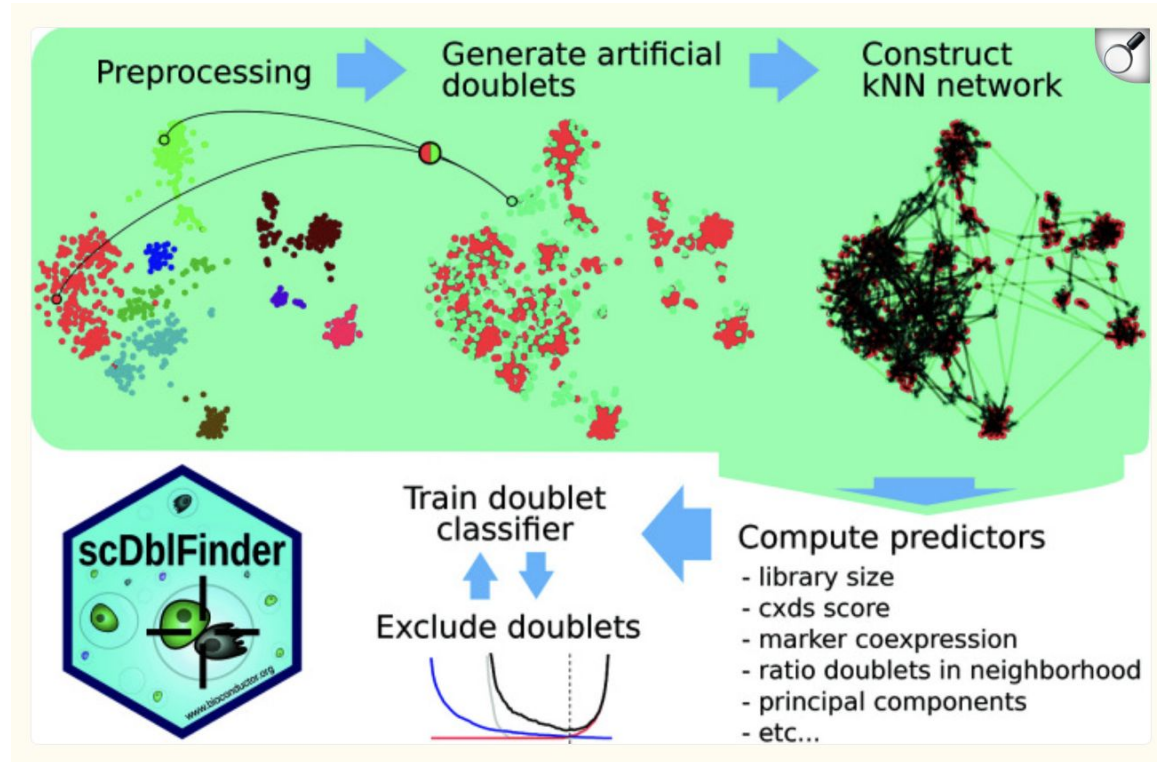


Homotypic doublet



scDblFinder: 1 sentence summary

- Generates realistic artificial doublets, summarizes real + synthetic cells, trains a classifier on these features to score each droplet, and then thresholds the scores to flag likely heterotypic doublets



scDbtFinder – words of advice from others

Doublet detection procedures should only be applied to libraries generated in the same experimental batch. It is obviously impossible for doublets to form between two cells that were captured separately. Thus, some understanding of the experimental design is required prior to the use of the above functions. This avoids unnecessary concerns about the validity of batch-specific clusters that cannot possibly consist of doublets.

It is also difficult to interpret doublet predictions in data containing cellular trajectories. By definition, cells in the middle of a trajectory are always intermediate between other cells and are liable to be incorrectly detected as doublets. Some protection is provided by the non-linear nature of many real trajectories, which reduces the risk of simulated doublets coinciding with real cells in

`computeDoubletDensity()`. One can also put more weight on the relative library sizes in `findDoubletClusters()` instead of relying solely on `N`, under the assumption that sudden spikes in RNA content are unlikely in a continuous biological process.

Start of the hands-on
portion